Aravind E S

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AI Software Engineer with 2 years of experience architecting and deploying production AI systems across cloud backends and edge devices, from hallucination-resistant agentic workflows (MCP, RAG) to real-time computer vision pipelines built on foundational models (YOLO, SAM 2.1, OCR/VLM), including a collision-warning system that achieved U.S. safety certification for production deployment.

Experience

Tata Elxsi — Dec 2024 – Present
AI Software Engineer — *Kerala, India*

Agentic AI & Compliance Platform

- Architected a production LLM agent enforcing read-only access, validated queries, and Pydantic schemas to prevent hallucinated responses on live operational data.
- Engineered a domain-grounded RAG layer fusing real-time database state with operating-procedure documentation for reliable, auditable agent reasoning.
- Implemented Redis caching and an asynchronous skill-discovery agent that mines usage patterns for reusable query skills without adding user-facing latency.

Computer Vision & Edge Deployment

- Deployed and optimized vision workloads (YOLO, SAM 2.1 zero-shot, OCR/VLM) on NVIDIA Jetson hardware, cutting inference latency by approximately 25% via TensorRT and CUDA.
- Delivered a collision-warning system that achieved U.S. safety certification for production deployment.
- Stabilized live-field video delivery by optimizing RTSP streaming pipelines (FFMPEG, GStreamer), improving stability by approximately 30% under degraded network conditions.
- Owned migration of the core data layer from SQLite to PostgreSQL and Azure PostgreSQL to support concurrent, high-throughput production workloads.

Tata Elxsi — Jan 2024 – Jun 2024
AI Software Developer Intern — *Kerala, India*

- Engineered radar point-cloud processing pipelines in ROS and extended pretrained YOLO architectures with dual detection heads to improve real-time detection recall.

Projects

Agentic AI Clinical Fall Intelligence Platform | *FastAPI, MCP, LLM Tool Calling, Docker* — 2025

- Engineered an agentic platform executing LLM-driven analysis on clinical fall-risk data via controlled, tool-based function calling (MCP) rather than unrestricted model access.
- Automated containerized production deployment on AWS EC2 using Nginx reverse proxying for isolated API routing.

Real-Time Edge Perception & Collision Warning System | *ONNX, TensorRT, CUDA, Jetson* — 2026

- Engineered an edge-deployed perception pipeline integrating camera ego-path detection with radar and GPS telemetry for real-time vehicle-risk analysis.
- Optimized TwinLiteNetPlus inference via ONNX and TensorRT/CUDA, enabling low-latency execution on NVIDIA Jetson edge hardware.

Technical Skills

Agentic AI: LLM Agents, MCP, LLM Tool Calling, RAG, Agentic Workflows, LangChain, LlamaIndex, Pydantic Schema Validation
Computer Vision: YOLO, SAM 2.1 (Zero-Shot), OCR/VLM, OpenCV, Multi-Camera Tracking, Camera Calibration
Edge & Inference: NVIDIA Jetson, TensorRT, CUDA, ONNX, Model Optimization
Backend & Infra: Python, C++, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, Celery, AWS EC2, AWS Bedrock, SageMaker, Nginx, Git

Certifications

- **AWS Certified AI Practitioner**
- **Rising Star Award** – Tata Elxsi

Education

Cochin University of Science and Technology — Kerala, India
Master of Computer Applications (MCA) — *2022 – 2024*

Mary Matha Arts and Science College — Kerala, India
Bachelor of Science in Physics — *2018 – 2021*